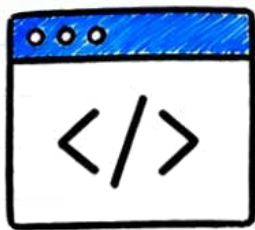


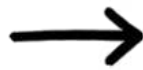


YOUR AWS DATA IS ENCRYPTED

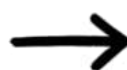
**BUT WHO PROTECTS
THE KEY?**



Application



Data 



Key



INTRODUCING

AWS KMS — KEY MANAGEMENT SERVICE

→ AWS KMS helps you create, manage and control the encryption keys used to protect your data.



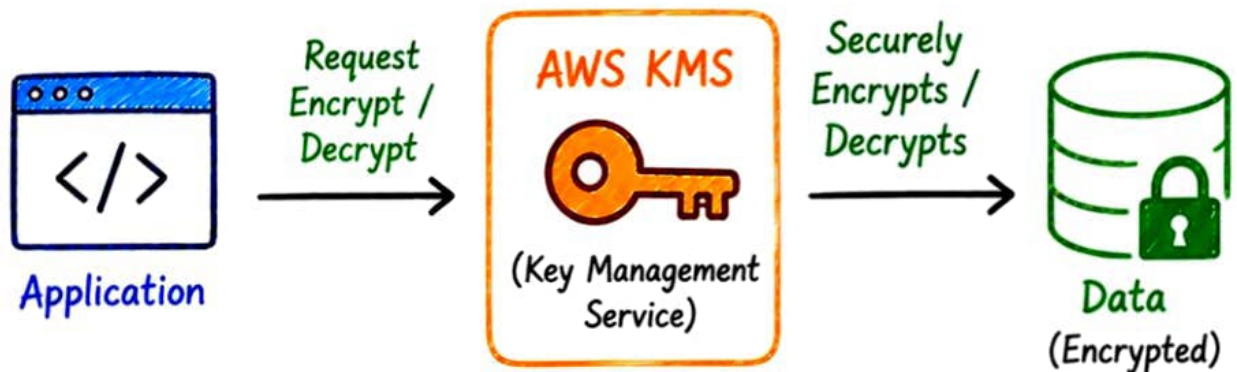
Encryption is only as secure
as the **key** protecting it.





WHAT IS AWS KMS?

AWS KMS (Key Management Service) is a managed service that helps you **create and control** the cryptographic keys used to **encrypt** and **decrypt** your data.



KEY CAPABILITIES



Centralized Key Management

Create, store and manage keys in one place.



Key Rotation

Automatically rotate keys to improve security.



Access Control

Use IAM and key policies to control who can use the keys.



Audit & Monitoring

All key usage is logged in Amazon CloudTrail.



AWS Service Integration

Seamlessly works with many AWS services.



AWS KMS gives you **control**, **security**, and **visibility** over your encryption keys 



ENVELOPE ENCRYPTION

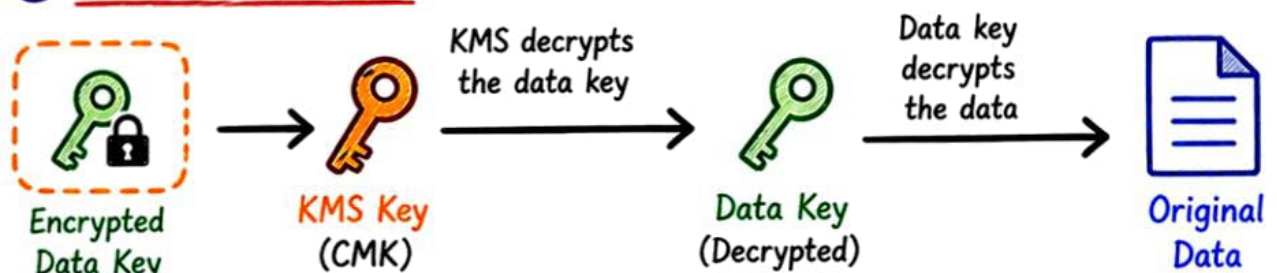
≧ THE IMPORTANT CONCEPT ≦

AWS KMS doesn't encrypt your data directly.
It protects a **data key**, and that key **encrypts** your data.

① ENCRYPTION FLOW



② DECRYPTION FLOW



KMS KEY PROTECTS THE **DATA KEY**.
DATA KEY PROTECTS THE **DATA**.



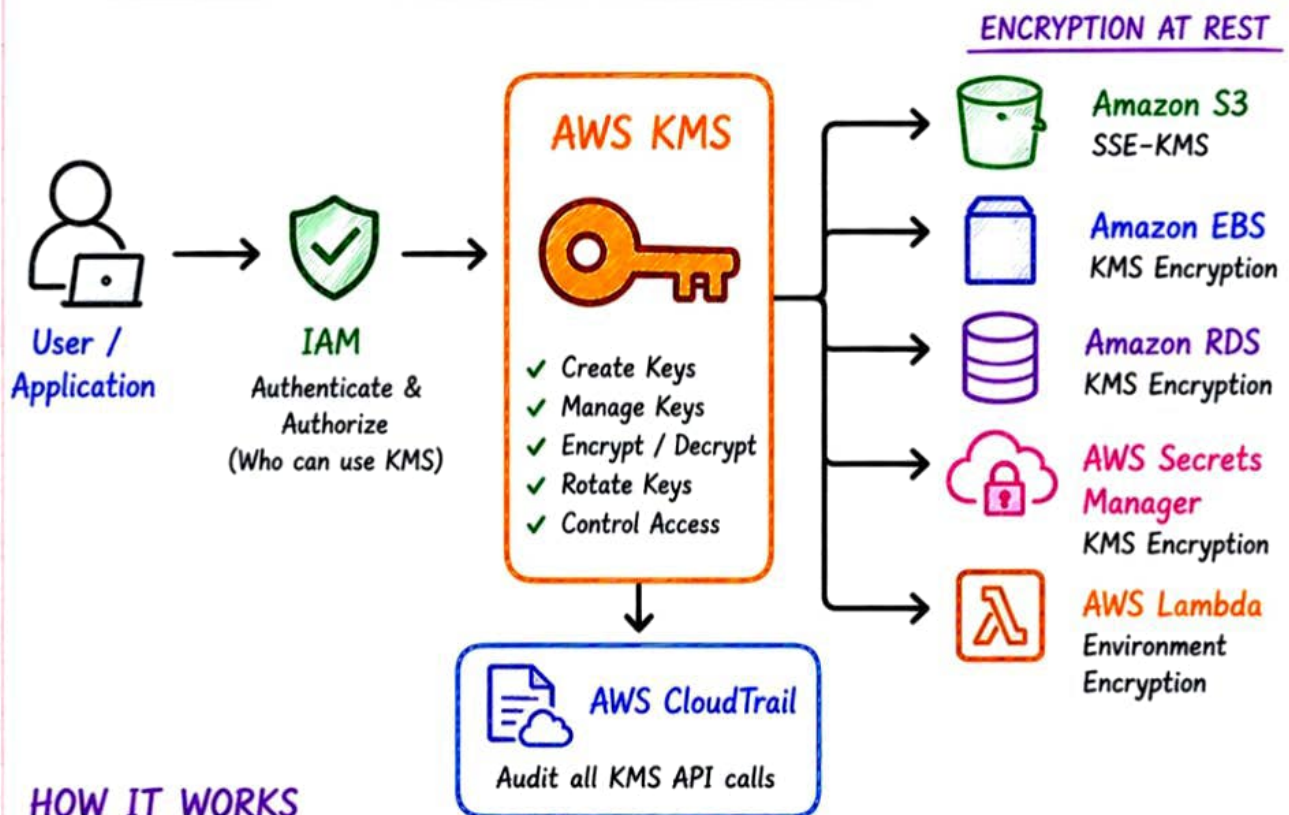
TAKEAWAY: If someone gets your encrypted data, without the **KMS key**, your data stays safe!





KMS IN REAL AWS ARCHITECTURE

AWS KMS is deeply integrated with many AWS services to **protect** your **data at rest**.



HOW IT WORKS



KMS CENTRALIZES KEY MANAGEMENT, ENSURES SECURITY, AND SIMPLIFIES ENCRYPTION ACROSS AWS SERVICES.



AWS MANAGED KEYS VS CUSTOMER MANAGED KEYS



Both types of keys use AWS KMS,
but they differ in **ownership**, **control** and **flexibility**.

 AWS MANAGED KEYS	 CUSTOMER MANAGED KEYS
- Created and managed by AWS. - Used by AWS services on your behalf.	 OWNERSHIP - Created, managed and owned by you. - You have full control over the key.
- AWS manages the key policy. - You have limited control over who can use the key.	 KEY POLICY & PERMISSIONS - You define the key policy. - Full control over who can use the key.
Automatic annual rotation (managed by AWS).	 ROTATION - You can enable automatic rotation (1 year – 10 years). - You can rotate manually when needed.
Cannot be scheduled for deletion.	 DELETION You can schedule key deletion (7–30 days).
Less visibility and control over key usage.	 VISIBILITY & CONTROL Full visibility in CloudTrail and full control over key usage.
- Good for most AWS services (S3, EBS, RDS, etc.). - Simpler, less management.	 USE CASES Use when you need strict control, compliance, or advanced security.



Need simplicity?

→ AWS Managed Key

Need control?

→ Customer Managed Key



For sensitive data, compliance and fine-grained access,
Customer Managed Keys are the better choice!



INTERVIEW CHEAT SHEET

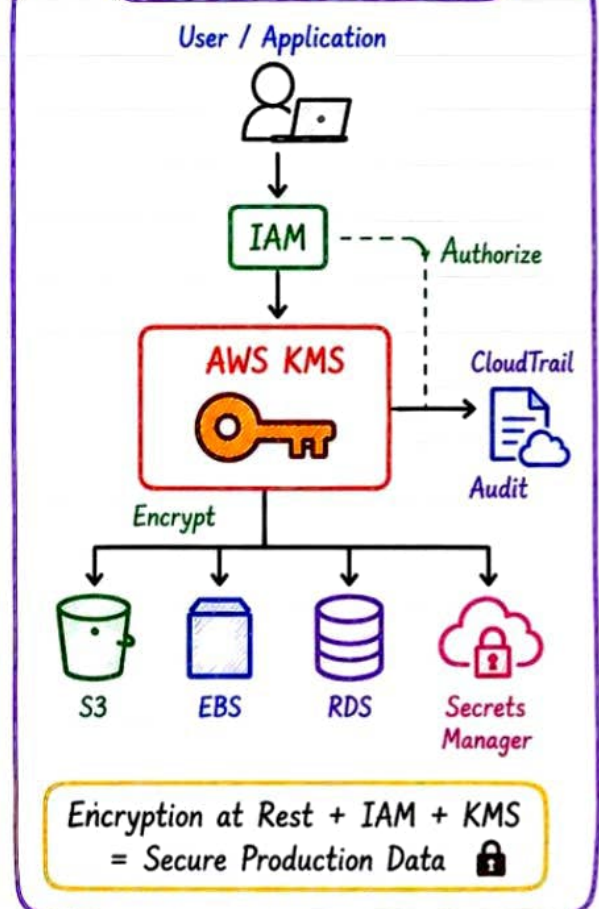
HOW WOULD YOU PROTECT SENSITIVE PRODUCTION DATA USING AWS KMS?



★ Strong answer framework - Step by Step

-  **Create / Choose KMS Key**
Create a customer managed KMS key or use an existing key.
-  **Define IAM & Key Policy**
Apply least-privilege IAM policies and key policies to control who can use the key.
-  **Enable Encryption on AWS Services**
Use the KMS key to enable encryption at rest on S3, EBS, RDS, Secrets Manager, etc.
-  **Use Envelope Encryption**
KMS protects the data key.
The data key encrypts the actual data.
-  **Audit with CloudTrail**
Logs all KMS API calls in CloudTrail for monitoring and compliance.
-  **Key Rotation & Lifecycle**
Enable automatic rotation (1-10 years) and define key deletion policy.

SAMPLE ARCHITECTURE



✗ COMMON MISTAKES

- ✗ Using AWS managed keys where more control is needed.
- ✗ Allowing * in key policy.
- ✗ Not enabling rotation.
- ✗ Not monitoring KMS usage.
- ✗ Storing secrets without using KMS.

KEY TAKEAWAY

IAM controls WHO 

KMS controls the KEY 

Encryption protects the DATA 



SAVE THIS FOR YOUR AWS INTERVIEW

